

AUTOMATED LINE-FOLLOWING ROBOT



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Line-following robots with pick-and-placement capabilities are commonly used in manufacturing plants. These move on a specified path to pick the components from specified locations and place them on desired locations.

Basically, a line-following robot is a self-operating robot that detects and follows a line drawn on the floor. The path to be taken is indicated by a white line on a black surface. The control system used must sense the line and manoeuvre the robot to stay on course while constantly correcting the wrong

moves using feedback mechanism, thus forming a simple yet effective closed-loop system.

Circuit description

Fig. 1 show the block diagram of the automated line-following robot. It consists of mainly four parts: two sensors, two comparators, one decision-making device and two motor drivers. The robot is built using microcontroller AT89C51 (used as the decision-making device), motor driver L293D, operational amplifier LM324 (comparator), phototransistor (sensor) and a few discrete components.

In the circuit, the sensors (photo-

transistors) are used to detect the white strip on a black background. The sensor output is fed to the microcontroller, which takes the decision and gives appropriate command to motor driver L293D so as to move the motor accordingly.

Sensor. The sensor senses the light reflected from the surface and feeds the output to the comparator. When the sensor is above the white background the light falling on it from the source reflects to the sensor, and when the sensor is above the black background the light from the source doesn't reflect to it. The sensor senses the reflected light to give an output, which is fed to the comparator.

Comparator. The comparator compares the analogue inputs from sensors with a fixed reference voltage. If this voltage is greater than the reference voltage the comparator outputs a low voltage, and if it is smaller the comparator generates a high voltage that acts as input for the decision-making device (microcontroller).

Microcontroller. The microcontrol-

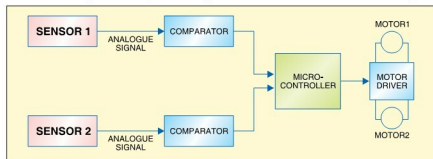


Fig. 1: Block diagram of automated line-following robot

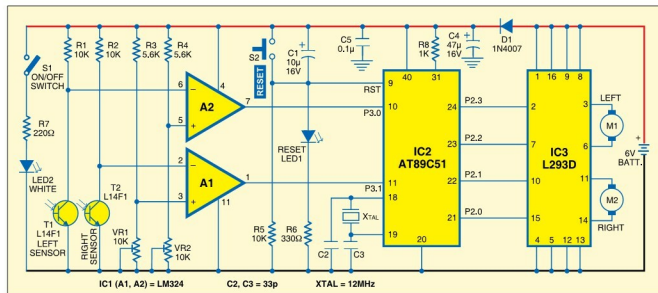


Fig. 2: Automated line-following robot circuit